

RIYA SHARMA

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PROFESSIONAL SUMMARY

Civil Engineering graduate with research experience in municipal solid waste characterization and resource recovery. My undergraduate research investigated the physical and chemical properties of municipal solid waste and evaluated the feasibility of material and energy recovery, including recycling, composting, and energy recovery. I have experience in field sampling, laboratory testing, data analysis, and technical research, and I am interested in advancing my research in sustainable waste management, waste-to-energy, and resource recovery.

RESEARCH INTERESTS

- Municipal Solid Waste Management
- Resource and Energy Recovery
- Waste Characterization and Treatment
- Sustainable Waste Management
- Circular Economy and Material Recovery
- Waste-to-Energy Systems

EDUCATION

Bachelor of Civil Engineering, National College of Engineering, Lalitpur, Nepal (2018–2023)
Percentage: 72.37%

Relevant Coursework: Structural Dynamics; Earthquake-Resistant Design; Steel Structures; Reinforced Concrete Structures; Structural Analysis; Finite Element Analysis; Engineering Mathematics; Soil Mechanics; Foundation Engineering; Engineering Geology; Engineering Hydrology; Fluid Mechanics

RESEARCH EXPERIENCE

Undergraduate Research- Municipal Solid Waste and Resource Recovery

National College of Engineering (2022–2023)

Undergraduate Project: Resource and Energy Recovery from Municipal Solid Waste – Lalitpur Metropolitan City

- Characterized municipal solid waste generated in Lalitpur Metropolitan City Ward 14 through household surveys, waste sampling, field measurements, laboratory analysis, and interviews.
- Determined waste generation rate, composition, density, moisture content, calorific value, pH, and NPK characteristics.
- Evaluated the feasibility of material and energy recovery from the municipal waste stream.

- Assessed recovery pathways including recycling, composting, and energy recovery, based on the physical and chemical characteristics of the waste.
- Found that organic waste constituted 81.55% of the MSW, with 53.24% moisture content and a heat content of 4.133 MJ/kg.
- Evaluated the suitability of the waste for energy recovery and found that the heat content was insufficient for practical incineration, while the high organic and NPK content indicated potential for composting and organic fertilizer production.
- Conducted quantitative analysis of recoverable materials, including plastics and paper, and assessed their potential for recycling and reuse.

PUBLICATION

Sharma, R., et al. (2023). *Characterizing Municipal Solid Waste and Exploring Resource Recovery Potential: A Case Study of Lalitpur Metropolitan City Ward 14*. Proceedings of the International Conference on Engineering and Technology (ICET 2023).

SELECTED ACADEMIC PROJECTS

Hydropower Potential Assessment Using Flow Duration and Power Duration Curves

- Developed flow duration curves from streamflow records.
- Generated power duration curves to evaluate hydropower generation potential.
- Performed hydrological data analysis using Microsoft Excel.
- Assessed river discharge variability for hydropower planning.

Seismic Performance Evaluation of a 3-Story Reinforced Concrete Frame Using ETABS

- Developed 3D ETABS model of RC moment-resisting frame.
- Applied gravity and lateral seismic loads based on code provisions.
- Performed modal analysis and response spectrum analysis.
- Evaluated interstory drift, base shear, and damage-sensitive response parameters.

RESEARCH & LABORATORY METHODS

- Municipal solid waste sampling and characterization
- Waste segregation and composition analysis
- Moisture-content and density determination
- Calorific-value assessment
- pH and NPK analysis
- Quantitative and statistical data analysis
- Field investigation and laboratory testing

PROFESSIONAL EXPERIENCE

Civil Engineer, Elysian Engineering Consultant & Building Pvt. Ltd. (2023–2024)
Structural drafting and modeling using ETABS and AutoCAD; assisted in load calculations and technical documentation.

Civil Engineer, Pathak Saurab JV & Pabita Nirman Sewa (2024–Present)

Supervised structural implementation, conducted inspections, and ensured quality compliance.

TECHNICAL SKILLS

Data Analysis: MS Excel, MATLAB

Structural: ETABS, SAP2000

Drafting: AutoCAD

Geospatial: ArcGIS, Google Earth

Programming: C

RESEARCH SKILLS

Literature Review • Data Analysis • Technical Writing • Field Investigation • Laboratory Investigation • Research Documentation

HONORS & ACHIEVEMENTS

- Excellence Award (Academic Scholarship), National College of Engineering (2023)
- General Registered Engineer – Nepal Engineering Council
- First Runner-Up – Inter-College Paper Bridge Competition

REFERENCES

Mr. Kamal Katwal

Senior Lecturer, Department of Civil Engineering

National College of Engineering, Nepal

Email: kamal@nce.edu.np

Relationship: Lecturer and Undergraduate Project Supervisor

Mr. Vimesh Paudel

Lecturer, Ara Institute of Canterbury, New Zealand

Email: vimesh.paudel@ara.ac.nz

Relationship: Former Lecturer

Mr. Shailendra Raj Giri

PhD Student, University of Newcastle, Australia

Email: sailendraraj.giri@uon.edu.au

Relationship: Undergraduate Project Supervisor